

# What Gets a Reply

Pre-registered observational study, n = 12,077 outreach emails. 2025 was used as the training set; Jan–Jul 2026 was held back as a sealed validation set and opened once.

## 0.1 · Objective

Measure two quantities and answer two questions.

|    | QUANTITY MEASURED                  | QUESTION                                       |
|----|------------------------------------|------------------------------------------------|
| Q1 | Reply rate to the first cold email | Which properties of the email predict a reply? |
| Q2 | Replies per follow-up touch        | How many follow-ups are productive?            |

Outcomes: **replied** = a human wrote back (label accuracy 99.0%); **interested** = the reply shows forward motion (92.8%). Every finding is reported on both.

## 0.2 · Principal results

| #  | RESULT                                                    | 2025         | 2026 VALIDATION | STATUS    |
|----|-----------------------------------------------------------|--------------|-----------------|-----------|
| R1 | Templated bodies collapse replies                         | 4.8% vs 9.3% | 1.8% vs 8.0%    | CONFIRMED |
| R2 | A stated reason-to-write-now doubles replies <sup>a</sup> | 7.9% vs 4.6% | 8.5% vs 4.1%    | CONFIRMED |
| R3 | Researched personalisation does not move replies          | +1.3 pp n.s. | +0.6 pp n.s.    | NULL      |
| R4 | Follow-up 1 is worth as much as the opener                | 5.0% vs 4.1% | 6.0% vs 5.8%    | HELD      |
| R5 | 2 of 19 predictions failed validation and are withdrawn   |              |                 | WITHDRAWN |

<sup>a</sup> Confirmed under the pre-committed rule (p=0.047); margin is fragile — 2 of 5 robustness specifications lift p above 0.05 (§7, L4). 2025 evidence is unambiguous (p<0.001 in every specification).

## 0.3 · Implications, graded by evidence

| ACTION                                                               | GRADE    |
|----------------------------------------------------------------------|----------|
| Stop sending identical bodies at scale from personal mailboxes       | STRONG   |
| Require every cold email to state its reason-to-write-now            | STRONG   |
| Always send follow-ups 1–2; usually 3; stop at 4 absent a new reason | MODERATE |
| Keep openers under ~100 words (no benefit below that)                | MODERATE |
| Stop research-heavy personalisation as a reply tactic                | MODERATE |
| Stop removing links; stop name-greeting rituals                      | MODERATE |

## 1 · Design

### 1.1 · Data

| STAGE                             | DEFINITION                                                                                                                    | N                         |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| Sent-email records                | All outbound engagement records, 4 internal domains                                                                           | ~117,000                  |
| Study frame                       | First email of a fresh push ( $\geq 30$ days silence), Gmail-recorded, commercial sender; bounces & existing threads excluded | 12,077                    |
| — cold pitches / event invites    | email type assigned blind by AI (2025 + 2026)                                                                                 | 5,153+1,286 / 3,179+1,837 |
| Reply candidates labelled         | human / auto-reply / calendar / bounce (99.0% accuracy)                                                                       | 16,695                    |
| Emails scored by blinded AI judge | 12 qualities × anchored rubric                                                                                                | 12,462                    |

### 1.2 · Design principles

- P1 — Within-sender comparison.** Every effect compares a rep's own emails with vs without a trait; rep skill and rep targeting cancel out.
- P2 — Rules before results.** Traits, thresholds, outcome definitions and pass/fail rules were committed to a timestamped record before any result was computed.
- P3 — Multiplicity control.** ~40 traits tested  $\Rightarrow$  1–2 false positives expected by chance; Benjamini–Hochberg correction applied within pre-declared families.
- P4 — Blind judging.** The AI judge saw subject + body only: sender, date, recipient identity and outcome redacted. Automated leak gate: 0 leaks / 12,462 items.
- P5 — Placebo.** The full analysis re-run on outcome-shuffled data  $\times 10$ : 5 spurious hits in 240 tests (2.1%), under the 5% ceiling. The machinery invents nothing.
- P6 — Training / validation split.** 2025 = training set (free exploration); Jan–Jul 2026 = validation set, sealed until every prediction was committed, then opened exactly once. A finding that fails validation is withdrawn, never rescued.

### 1.3 · Two measurement layers

| LAYER | WHAT             | HOW MEASURED                                                       | EXAMPLES                                                    |
|-------|------------------|--------------------------------------------------------------------|-------------------------------------------------------------|
| L1    | Countable traits | Audited code (61 regression tests)                                 | length, questions, links, subject shape, template detection |
| L2    | Judged qualities | Blinded AI, anchored 1–5 rubric; dual-model reliability check (§5) | why-now, research signal, bespokeness, tone                 |

### 1.4 · Verdict vocabulary (fixed before the validation set was opened)

| VERDICT             | DEFINITION                                                                                        |
|---------------------|---------------------------------------------------------------------------------------------------|
| <b>CONFIRMED</b>    | Significant in 2025 after correction; same sign and $p < 0.05$ in the 2026 validation set.        |
| <b>HELD</b>         | A committed non-effect-size prediction (curve shape, ranking, band) satisfied in 2026.            |
| <b>DIRECTION</b>    | Same sign in 2026 but not significant there (2026 is $\frac{1}{4}$ the size of 2025).             |
| <b>NULL</b>         | No significant effect in either year, with sample sizes able to detect effects the size of R1/R2. |
| <b>INCONCLUSIVE</b> | 2026 too noisy to rule either way under the pre-committed equivalence bound.                      |
| <b>FAILED</b>       | 2026 estimate contradicts 2025 and statistically excludes the 2025 effect size. Withdrawn.        |
| <b>UNTESTABLE</b>   | The trait effectively vanished from 2026 email; no test possible.                                 |

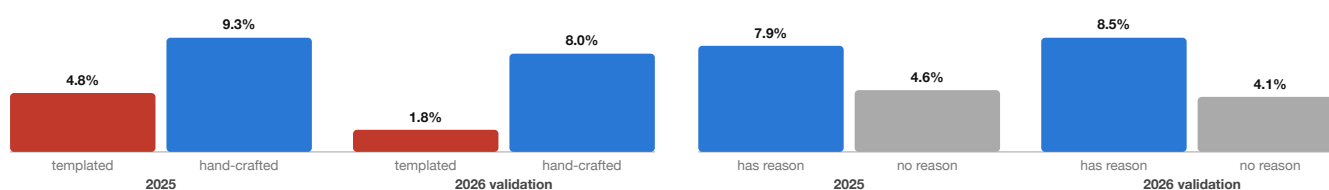
## 2 · Q1 findings

All effects: percentage points (pp) on reply rate, within-sender, cold pitches unless marked. 2025 n=5,153 (base 6.9%); validation 2026 n=1,286 (base 6.5%). The companion "interested" outcome agrees in sign throughout.

### 2.1 · Confirmed effects

| TRAIT                                               | 2025 EFFECT (PP) · TRAINING | 2026 EFFECT (PP) · VALIDATION | 2026 P | VERDICT   |
|-----------------------------------------------------|-----------------------------|-------------------------------|--------|-----------|
| Templated body (3+ identical copies)                | -5.4                        | -3.8                          | 0.005  | CONFIRMED |
| States a reason to write now (why-now) <sup>a</sup> | +4.6                        | +2.9                          | 0.047  | CONFIRMED |

<sup>a</sup> Fragile 2026 margin: main specification p=0.047; 2 of 5 robustness specifications lift p to 0.10 / 0.07. Sign holds in all five; the fair-window sensitivity strengthens it (+3.3, p=0.024). See §7, L4.



R1 — reply rate, templated vs hand-crafted

R2 — reply rate, with vs without a stated reason

R1 fingerprints: bold text (−4.3 pp) and mid-body name reuse (−5.4 pp) looked harmful in 2025, but 77–80% of emails carrying them were templates — one phenomenon, counted once.

### 2.2 · Directional effects (sign replicated; 2026 underpowered)

| TRAIT                                      | 2025 EFFECT (PP) · TRAINING | 2026 EFFECT (PP) · VALIDATION | 2026 P | VERDICT   |
|--------------------------------------------|-----------------------------|-------------------------------|--------|-----------|
| Body ≤100 words <sup>b</sup>               | +4.5                        | +0.8                          | 0.641  | DIRECTION |
| Subject line is a question <sup>c</sup>    | +4.5                        | +1.7                          | 0.565  | DIRECTION |
| Asks 2+ questions                          | -2.5                        | -2.5                          | 0.216  | DIRECTION |
| Event invite: templated body               | -10.1                       | -2.3                          | 0.211  | DIRECTION |
| Event invite: body ≤100 words <sup>b</sup> | +17.5                       | +1.3                          | 0.506  | DIRECTION |

<sup>b</sup> 2026 excludes the 2025 effect size — direction survives, magnitude does not. <sup>c</sup>

Question-subjects fell from 19% of 2025 cold pitches to 6% in 2026 (n=79); no power. Event-invite rows: 2025 n=3,179 (base 15.1%), 2026 n=1,837 (13.5%).

### 2.3 · Null effects — equal standing with the positives

| TRAIT                                                  | 2025 EFFECT (PP) · TRAINING | 2026 EFFECT (PP) · VALIDATION | 2026 P | VERDICT |
|--------------------------------------------------------|-----------------------------|-------------------------------|--------|---------|
| Cites researched facts about recipient <sup>d</sup>    | +1.3                        | +0.6                          | 0.753  | NULL    |
| Reads written-only-for-them (bespokeness) <sup>d</sup> | +1.9                        | +2.1                          | 0.473  | NULL    |
| Single clear, specified ask <sup>d</sup>               | +0.0                        | +0.9                          | 0.653  | NULL    |
| Contains a link                                        | +4.9                        | +1.5                          | 0.547  | NULL    |
| Greets recipient by name                               | -2.1                        | +2.1                          | 0.190  | NULL    |
| Mentions recipient's company                           | +0.0                        | -0.4                          | 0.862  | NULL    |
| Polished writing                                       | +1.5                        | +1.4                          | 0.314  | NULL    |

<sup>d</sup> The three personalisation nulls are pre-registered primary dimensions; under the pre-committed equivalence bound the 2026 CIs cannot certify the effect is  $\leq 3$  pp, so formally "still nothing, twice" rather than "proven zero". Research-signal and bespokeness correlate  $r=0.83$  — one finding, not two.

## 2.4 · Failed, inconclusive, untestable

| TRAIT                                         | 2025 EFFECT (PP) · TRAINING | 2026 EFFECT (PP) · VALIDATION | 2026 P | VERDICT      |
|-----------------------------------------------|-----------------------------|-------------------------------|--------|--------------|
| Recipient-focused writing                     | +5.3                        | -0.4                          | 0.851  | FAILED       |
| Event invite: contains bullet points          | -9.6                        | +1.3                          | 0.730  | FAILED       |
| Contains bold text <sup>e</sup>               | -4.3                        | +5.7                          | 0.347  | INCONCLUSIVE |
| Reuses recipient's name mid-body <sup>f</sup> | -5.4                        | —                             | —      | UNTESTABLE   |

<sup>e</sup> Bold-text cell shrank to  $n=126$ ; wrong sign inside noise — the pre-committed rule scores this inconclusive, not refuted. <sup>f</sup> Mid-body name reuse fell from 7.7% of 2025 cold pitches to 2 emails in 2026; the habit itself disappeared.

## 2.5 · Word-length shape (2025 cold pitches)

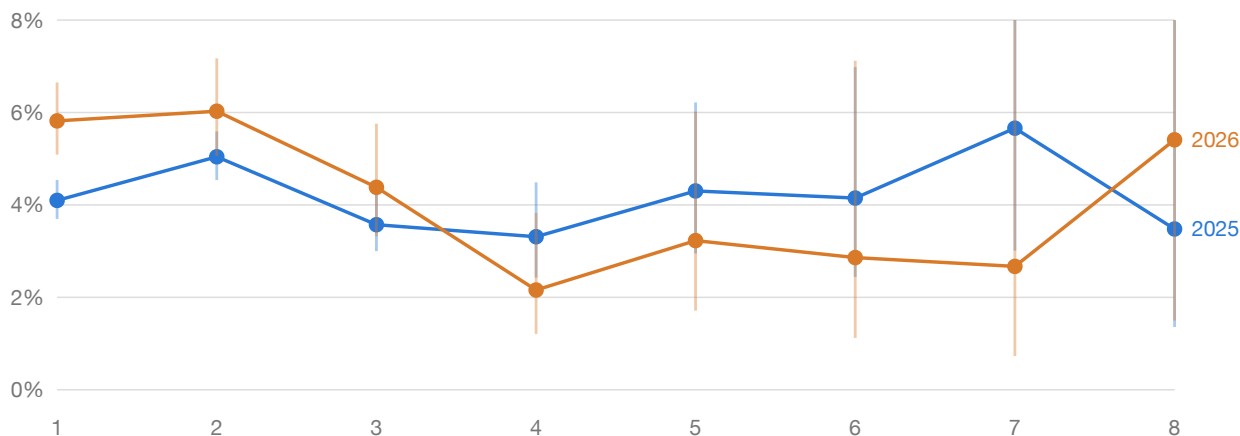
| WORDS   | N     | REPLY RATE |
|---------|-------|------------|
| <50     | 163   | 11.0%      |
| 50–99   | 1,205 | 10.2%      |
| 100–149 | 3,213 | 5.5%       |
| 150–249 | 571   | 6.7%       |

The benefit is a **floor at ~100 words**, not a slope: below 100 there is no further gain ( $\leq 60$ -word emails do no better than 61–100-word emails).

### 3 · Q2 findings

| TOUCH | RECEIVED IT (2025) | REPLY RATE (2025) | CUMULATIVE (2025) | RECEIVED IT (2026) | REPLY RATE (2026) |
|-------|--------------------|-------------------|-------------------|--------------------|-------------------|
| 1     | 8,591              | 4.1%              | 4.1%              | 3,486              | 5.8%              |
| 2     | 6,630              | 5.0%              | 8.0%              | 1,974              | 6.0%              |
| 3     | 3,505              | 3.6%              | 9.4%              | 1,095              | 4.4%              |
| 4     | 1,178              | 3.3%              | 9.9%              | 509                | 2.2%              |
| 5     | 605                | 4.3%              | 10.2%             | 279                | 3.2%              |
| 6     | 313                | 4.2%              | 10.3%             | 140                | 2.9%              |
| 7     | 159                | 5.7%              | 10.5%             | 75                 | 2.7%              |
| 8     | 115                | 3.5%              | 10.5%             | 37                 | 5.4%              |

Reply rate = share of still-silent recipients who replied after that touch. All email types; 8,591 pushes (2025), 3,486 (2026). Late-touch 2026 cells are small — see uncertainty bars in the chart.



#### 3.1 · Statements

- Follow-up 1 out-performs the opener in both years (5.0 vs 4.1; 6.0 vs 5.8). Predicted; **HELD**.
- 61% of all replies arrive after the opener; ~90% of eventual replies are in by touch 3.
- Touches 5–8 added 0.6 pp combined (2025). Predicted ≥85% of replies by touch 4; observed 96% (2026); **HELD**.
- Constraint: who receives touch 5 is the rep's choice — late-touch rates describe survivors.

#### SECTION 4 · CORPUS PROFILE

### 4 · What the emails look like (blind ratings; outcomes not attached)

| PROPERTY                                       | SHARE |
|------------------------------------------------|-------|
| states a real “why now”                        | 80.3% |
| well-polished (4–5)                            | 81.4% |
| named proof from their field (4–5)             | 48.8% |
| visible mail-merge or near it (bespokeness ≤2) | 44.2% |
| no pain hypothesis at all (score 1)            | 60.4% |
| mostly about the recipient (4–5)               | 8.7%  |

The corpus is weakest exactly where the two confirmed effects live: 20% of emails state no reason-to-write, and 44% are visible mail-merge. Writing quality (polish, clear asks) is already high and is not the constraint.

## 5 · Reliability of the AI judge

Reliability = agreement beyond chance (Cohen's  $\kappa$ ) on the exact cut used in analysis; 0 = coin flip, 1 = perfect. Self = same model re-scoring 1,246 emails; second AI = independent model, 1,000 emails.

| JUDGED QUALITY        | K SELF | K SECOND AI | STATUS    |
|-----------------------|--------|-------------|-----------|
| Reason to write now   | 0.72   | 0.72        | RETAINED  |
| Research on recipient | 0.75   | 0.73        | RETAINED  |
| Pain hypothesis       | 0.69   | 0.64        | RETAINED  |
| Value specificity     | 0.67   | 0.62        | RETAINED  |
| Customer proof        | 0.65   | 0.66        | RETAINED  |
| Bespokeness           | 0.56   | 0.60        | RETAINED  |
| Ask clarity           | 0.42   | 0.57        | RETAINED  |
| Polish                | 0.43   | 0.46        | RETAINED  |
| Recipient focus       | 0.43   | 0.45        | RETAINED  |
| Peer tone             | 0.32   | 0.32        | DISCARDED |
| Economy               | 0.22   | 0.26        | DISCARDED |

1. Mean gap between self- and cross-model agreement:  $-0.02 \Rightarrow$  the two discarded qualities are ambiguous *questions*, not a weak model. Nothing in this document rests on them.
2. Halo check: the 12 qualities are separable (mean pairwise  $|r| = 0.19$  vs 0.60 threshold; first component 28% vs 50%). Exception: research-signal ~ bespokeess  $r = 0.83 \Rightarrow$  one finding.
3. Confound check: judged "economy" tracks word count ( $r = -0.48$ )  $\Rightarrow$  treated as a restatement of L1 length; judged proof tracks length ( $r = +0.42$ )  $\Rightarrow$  its negative effect disappears under a length control and is not reported as a finding.

## 6 · All 19 predictions

| PREDICTION                                        | RULE             | VERDICT                |
|---------------------------------------------------|------------------|------------------------|
| Templated bodies reply less                       | sign + $p < .05$ | CONFIRMED              |
| Reason-to-write-now replies more (both outcomes)  | sign + $p < .05$ | CONFIRMED <sup>a</sup> |
| Its effect lands in +1.6...+7.6 pp                | band             | HELD                   |
| It ranks #1 among judged qualities                | rank             | HELD                   |
| Follow-up 1 $\approx$ opener ( $\pm 2$ pp)        | band             | HELD                   |
| $\geq 85\%$ of replies by touch 4                 | threshold        | HELD                   |
| "Interested" mirrors "replied" on R1-class traits | sign $\times 3$  | HELD                   |
| Pain/proof negatives vanish under length control  | n.s. $\times 2$  | HELD                   |
| Discarded qualities produce nothing               | n.s. $\times 2$  | HELD                   |
| Body $\leq 100$ words replies more                | direction only   | DIRECTION              |
| Question subject replies more                     | direction only   | DIRECTION              |
| 2+ questions replies less                         | direction only   | DIRECTION              |
| Invites: templated replies less                   | direction        | DIRECTION              |

|                                        |                   |                     |
|----------------------------------------|-------------------|---------------------|
| Invites: short replies more            | direction         | <b>DIRECTION</b>    |
| Bold text replies less                 | direction only    | <b>INCONCLUSIVE</b> |
| Personalisation trio stays ≤3 pp       | equivalence bound | <b>INCONCLUSIVE</b> |
| Recipient-focused writing replies more | sign              | <b>FAILED</b>       |
| Invites: bullet points reply less      | sign              | <b>FAILED</b>       |
| Mid-body name reuse replies less       | —                 | <b>UNTESTABLE</b>   |

## 7 · What this study cannot say

1. **L1 — Scope.** Gmail-recorded outbound only (hand-sent + Apollo). Amplemarket sends have no reliable reply trail and are excluded; they grew from 20% of outreach (2025) to 71% (2026). Findings cover ~75% of 2025 outbound but <30% of 2026's, and say nothing about the Amplemarket channel.
2. **L2 — Sending tool.** Tool-sent email replies ~2 pp lower, but the offset is flat: 2 of 29 trait-effects differ by route (the chance rate). It shifts baselines, not the recipe; likely partly a tracking artefact. No claim made.
3. **L3 — Causality.** Within-sender comparison removes rep-level differences, not within-rep targeting (a rep may save better emails for better prospects). Effect sizes are upper bounds; directions are the trustworthy part. R1/R2 are the natural A/B-test candidates.
4. **L4 — R2's 2026 margin.** Confirmed at  $p=0.047$  under the pre-committed rule; two of five robustness specifications lift  $p$  above 0.05 (sign always holds; the fair-window outcome strengthens it to  $p=0.024$ ). Reported as confirmed with a fragile margin.
5. **L5 — Power.** 2026 is  $\frac{1}{4}$  of 2025; four verdicts are inconclusive by design honesty, not evidence of absence.
6. **L6 — Known residuals.** 155 Amplemarket emails in scope on a technicality (all analyses re-run without them; nothing changes); 4 test emails in frame (0.03%); 55 dedup boundary pairs of 224,756 touches; ~0.1% replies missed via colleague hand-offs; 0.5% non-English emails judged with an English rubric; July 2026 reply windows clipped (97% of replies arrive  $\leq 30$  days; the fair-window sensitivity strengthens both confirmations).

### 7.1 · Traceability

Pre-registration and correction log: rules/RUN2\_PREREGISTRATION.md (§1–§9.9).  
 Findings and predictions: docs/13–14. Pre-hold-out audit: docs/15. Hold-out verdicts: docs/16. Scientific audit: docs/17. Every figure in this document is generated from those committed files by scripts/build\_exec\_doc.py; none is typed by hand.